

Pseudodifferential operators, boundary value problems, and layer potentials on manifolds with cylindrical ends

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Dedicated to the memory of Professor Gabriela Kohr

Joint work with **Victor Nistor** (Metz) and **W.L. Wendland** (Stuttgart)

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- 9 Well-posedness of the Dirichlet problem for the Stokes system

Preliminaries

- We study the **Fredholm, invertibility, and the structure of the (pseudo)-inverse of the generalized Stokes operator**

$$\Xi_{V,V_0} = \begin{pmatrix} \mathbf{L} + V & \nabla \\ \nabla^* & -V_0 \end{pmatrix} = \begin{pmatrix} \mathbf{L} & \nabla \\ \nabla^* & 0 \end{pmatrix} + \begin{pmatrix} V & 0 \\ 0 & -V_0 \end{pmatrix}$$

under some non-negativity assumptions of the smooth potentials V and V_0 on **manifolds with straight cylindrical ends**, where $\mathbf{L} := 2\text{Def}^*\text{Def}$ is the *deformation Laplacian*.

- We define the **Stokes layer potentials S and $\frac{1}{2} + K$** on a domain with straight cylindrical ends and show that they are **pseudodifferential operators**, and study their **Fredholm and invertibility properties**.
- Applications to BVPs for the generalized Stokes and NS systems.
- We develop an “**algebra tool kit**” to deal with **limit and jump relations of layer potential operators**.
- **M. Mitrea** and **V. Nistor** (2007): the Laplace operator; **Mitrea** and **Taylor** method on compact manifolds.

Extensions

Many results and extensions for domains $\Omega \subset \mathbb{R}^n$:

A. Cialdea, S. Chandler-Wilde, M. Costabel, M. Dalla Riva, M. Dauge, C. Kenig, M. Lanza de Cristoforis, V. Mazya, S.E. Mikhailov, D. & I. & M. Mitrea, P. Musolino, D. Natroshvili, V. Nistor, G. Verchota, W.L. Wendland, ...

We want to extend these results to the case when Ω is a **domain with cylindrical ends** (inside some manifold). (A more complicated **operator algebra**.)

Extensions to manifolds: D. & I. & M. Mitrea, V. Nistor, M. Taylor.

Wave-guides: Bonnet-Ben Dhia, Bonnerier, Chesnel, Niclas, Krejcirik, ...

The Mitrea-Taylor method

Let us assume that $\Omega \subset M$ = a compact, smooth manifold.

The first issue: $\Delta := d^*d$ may no longer be invertible on M .

The **Mitrea-Taylor method** is:

- to replace Δ with $\Delta + V$, where V is a suitable potential, and
- to prove the invertibility of $\Delta + V$ and use $(\Delta + V)^{-1}$.
- to set

$$Sf(x) = (\Delta + V)^{-1}(f \otimes \delta_{\partial\Omega}) \quad \text{and} \quad Df(x) = (\Delta + V)^{-1}(f \otimes \delta'_{\partial\Omega}).$$

To define the **single and double layer potential operators** S and D :

- $\langle g \otimes \delta_{\partial\Omega}, h \rangle := \int_{\partial\Omega} h(x)g(x)d\sigma(x)$, $\langle g \otimes \delta'_{\partial\Omega}, h \rangle := \int_{\partial\Omega} \partial_\nu h g d\sigma$,
- $d\sigma$ is the surface measure on $\partial\Omega$.

Then define and invert the **boundary versions** of S and D : S and $\frac{1}{2}\mathbb{I} + K$.

The method of layer potentials ($\mathcal{S}$)

The definition of $\mathcal{S}$:

$$\mathcal{S}g := (\Delta + V)^{-1}(g \otimes \delta_{\partial\Omega}).$$

This definition gives that (for $x \in \Omega$)

$$(\Delta + V)\mathcal{S}g = (\Delta + V)((\Delta + V)^{-1}(g \otimes \delta_{\partial\Omega})) = g \otimes \delta_{\partial\Omega} = 0.$$

To solve the Dirichlet problem: **match the boundary conditions.**

Let $x \rightarrow z \in \partial\Omega$. This gives the **single layer potential operator**

$$\mathbf{S}g(z) := (\mathcal{S}g)|_{\partial\Omega}(z), \quad z \in \partial\Omega.$$

Summary of the single layer potential method

Let $g = \mathbf{S}^{-1}f$. Then $u := \mathcal{S}g$ solves the Dirichlet problem

$$(\Delta + V)u = 0 \text{ in } \Omega, \quad u = f \text{ on } \partial\Omega.$$

Ellipticity and the classical Fredholm criteria on compact manifolds

- $P = \text{order } m$ (classical, pseudo)differential operator on a compact manifold M , and hence $P : H^s(M) \rightarrow H^{s-m}(M)$ is **continuous**.
- P is **elliptic** if its principal symbol $\sigma_m(P)$ is invertible outside 0.

Theorem (classical result)

Assume that M is **compact**. Let $P \in \Psi^m(M)$ be a **classical pseudodifferential operator of order m** . Then, for any $s \in \mathbb{R}$,

$P : H^s(M) \rightarrow H^{s-m}(M)$ is **Fredholm** if and only if P is **elliptic**.

- **Domains with cylindrical ends: more complex or different issues:**
 - **The Fredholm condition(s)** (the classical one is **not true**) and
 - The L^2 -unique continuation property.
- Extension of the method to **manifolds with (straight) cylindrical ends**.

Extension of the method on manifolds with cylindrical ends

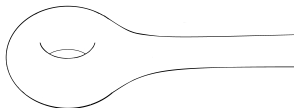


Figure: Manifold with straight cylindrical ends.

N has a boundary and M does not (both have straight cylindrical ends).

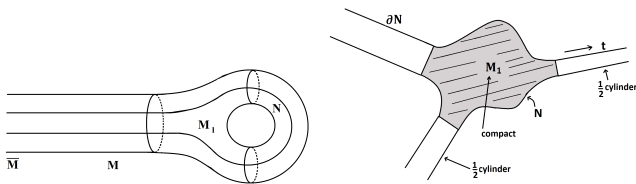


Figure: Manifolds with boundary (or domains) with cylindrical end

Manifolds with cylindrical ends

- M_0 = a smooth, compact Riemannian manifold, $\partial M_0 \neq \emptyset$.
- In a tubular neighborhood $V \cong \partial M_0 \times [0, 1)$ of the boundary ∂M_0 in M we consider a metric of product type $g_{\partial M_0}(x) + (dt)^2$.
- We endow the cylindrical end $\partial M_0 \times (-\infty, 0]$, $t \in (-\infty, 0]$, again with a product-type metric $g_{\partial M_0}(x) + (dt)^2$.

Definition

The manifold

$$M := M_0 \cup (\partial M_0 \times (-\infty, 0]),$$

with the induced metric, is called a **manifold with straight cylindrical ends** ($\partial M_0 \equiv \partial M_0 \times \{0\}$.)

Main result

Theorem (MK, Nistor, Wendland, 2026)

Let $\Omega \subset M$ be a domain with straight cylindrical ends and let $V, V_0 \geq 0$ be smooth translation invariant and non-negative at infinity. The operator

$$\Xi_{V, V_0} = \begin{pmatrix} \mathbf{L} + V & \nabla \\ \nabla^* & -V_0 \end{pmatrix} = \begin{pmatrix} \mathbf{L} & \nabla \\ \nabla^* & 0 \end{pmatrix} + \begin{pmatrix} V & 0 \\ 0 & -V_0 \end{pmatrix}$$

is **invertible**, and hence the single and double layer potential operators $\mathcal{S}$ and $\mathbf{K}$ associated to Ξ_{V, V_0} are defined. Moreover, $\mathcal{S}$ and $\frac{1}{2}\mathbb{I} + \mathbf{K}$ are **invertible**, and hence the Dirichlet problem on Ω is well posed.

We need to consider **suitable classes of (pseudodifferential) operators** that take into account the (translation invariant) structure of Ξ_{V, V_0} on manifolds with cylindrical ends.

Operators on manifolds with cylindrical ends

- d = exterior derivative operator on M .
- $\nabla := \sharp \circ d : C^\infty(M) \rightarrow C^\infty(M; TM)$ the gradient operator.
Recall that $\sharp : T^*M \rightarrow TM$ is the isomorphism defined by the Riemannian metric g of M (the “musical isomorphism”).
- $\Delta = \Delta_M = d^*d$ = **Laplace-Beltrami operator** on M .

Then

$$\Delta = \Delta_M = -\partial_t^2 + \Delta_{\partial M_0} \text{ on cylindrical end } \partial M_0 \times (-\infty, 0],$$

- Def = the **deformation operator** on M .
- **Translation invariant** (in a neighborhood of infinity) **operators**.

Translation invariant operators near infinity

Let $\Phi_s : \partial M_0 \times (-\infty, 0] \rightarrow \partial M_0 \times (-\infty, -s]$, $s \geq 0$, denote the isometry given by translation with $-s$ in the t -direction, that is

$$\Phi_s(x, t) = (x, t - s).$$

Definition

A linear continuous operator $P : C_0^\infty(M) \rightarrow C_0^\infty(M)'$ is called a **translation invariant at (in a neighborhood of) infinity** if

- its distribution kernel is supported in a neighborhood of the diagonal $\{(x, y) \in M \times M : \text{dist}(x, y) < \varepsilon\}$ for some $\varepsilon > 0$, and
- there exists $R > 0$ such that

$$P\Phi_s(f) = \Phi_s P(f), \quad \forall f \in C_0^\infty(\partial M_0 \times (-\infty, -R)), \quad \forall s > 0.$$

Thus, $P(x, t, x', t') = P(x, t - s, x', t' - s)$, $s > 0$.

Translation invariant operators near infinity

Next, we use the notation

- $\Psi_{\text{inv}}^m(M)$ = the space of classical pseudodiff. operators P of order $\leq m$ that are **translation invariant in a neighborhood of infinity**.
- $\Psi_{\text{inv}}^\infty(M) := \bigcup_{m \in \mathbb{Z}} \Psi_{\text{inv}}^m(M)$, $\Psi_{\text{inv}}^{-\infty}(M) := \bigcap_{m \in \mathbb{Z}} \Psi_{\text{inv}}^m(M)$.
- Similar, for *vector bundles* $E, F \rightarrow M$,

$$\Psi_{\text{inv}}^m(M; E, F), \quad \Psi_{\text{inv}}^\infty(M; E, F), \quad \Psi_{\text{inv}}^{-\infty}(M; E, F).$$

If $E = F$: $\Psi_{\text{inv}}^m(M; E)$.

Lemma

If M is a manifold with cylindrical ends, then any $P \in \Psi_{\text{inv}}^m(M; E)$ extends to a continuous operator $P : H^{m'}(M; E) \rightarrow H^{m'-m}(M; E)$, $m' \in \mathbb{R}$.

Essentially translation invariant operators

$\Psi_{\text{inv}}^{\infty}(M)$ is **not stable under inversion/holomorphic functional calculus**.

Definition

$\Psi_{\text{ess}}^{\infty}(M)$ = **operators essentially translation invariant**:

- $\Psi_{\text{ess}}^m(M) := \Psi_{\text{inv}}^m(M) + \Psi_{\text{ess}}^{-\infty}(M)$.
- $\Psi_{\text{ess}}^{-\infty}(M)$ = **suitable closure** of $\Psi_{\text{inv}}^{-\infty}(M)$ (w.r.t the norms $\|\cdot\|_{m',m}$ of linear operators $H^{m'}(M) \rightarrow H^m(M)$, $m', m \in 2\mathbb{Z}$, **and** with respect to the norms $\|\text{ad}_x^k(P)\|_{m',m}$, $k \in \mathbb{Z}_+$.)
- Similar for $\Psi_{\text{ess}}^m(M; E, F)$ if $E, F \rightarrow M$ are given vector bundles.

- $\Psi_{\text{ess}}^{\infty}(M)$ is larger than $\Psi_{\text{inv}}^{\infty}(M)$.
- $\Psi_{\text{inv}}^{\infty}(M)$ is a subset of the **b-calculus** introduced/studied by Melrose, Mazzeo and Melrose, Schulze, Schrohe, Grieser, Bierstone and Milman, Beals, and others, but $\Psi_{\text{ess}}^{\infty}(M)$ is not.
- Our algebra is **stable under inversion/holomorphic functional calculus**.

Theorem (Mitreă-Nistor, MK-Nistor-Wendland, 2024)

Let $T \in \Psi_{\text{ess}}^m(M; E, F)$, $m \geq 0$, be such that T is **invertible** as a (possibly unbounded) operator on $L^2(M; E)$. If $m > 0$, we assume also that T is **elliptic**. Then $T^{-1} \in \Psi_{\text{ess}}^{-m}(M; F, E)$.

Specific feature: **Limit operator**

Definition

Let $P \in \Psi_{\text{ess}}^\infty(M)$. The **limit operator** associated to P is the unique $\mathbb{R}$ -invariant (i.e. translation invariant) order $\leq m$ pseudodifferential operator

$\tilde{P} : C_0^\infty(\partial M_0 \times \mathbb{R}) \rightarrow C_0^\infty(\partial M_0 \times \mathbb{R})$ s.t.

$$k_{\tilde{P}}(t, x, t', x') := \lim_{s \rightarrow \infty} k_P(t - s, x, t' - s, x').$$

- $\tilde{P}$ is **well defined** and **translation invariant**.
- Thus, for $P \in \Psi_{\text{inv}}^\infty(M)$, define the **limit operator** $\mathcal{R}_\infty(P) := \tilde{P}$,

$$\mathcal{R}_\infty(P)(u) := \lim_{s \rightarrow \infty} \Phi_{-s} [P \Phi_s(u)|_{\{t < 0\}}],$$

where Φ_s is the translation by s on the cylinder $\partial M_0 \times \mathbb{R}$.

Theorem (Kondratiev, Melrose and Mendoza)

Let $M = M_0 \cup (\partial M_0 \times (-\infty, 0])$ be a manifold with cylindrical ends and $P \in \Psi_{\text{ess}}^m(M; E, F)$. Then $P : H^s(M; E) \rightarrow H^{s-m}(M; F)$ is a **Fredholm operator** if and only if

- it is **elliptic** and
- the **limit operator**

$$\tilde{P} : H^s(\partial M_0 \times \mathbb{R}; E) \rightarrow H^{s-m}(\partial M_0 \times \mathbb{R}; F)$$

is invertible.

Proof: using **operator theory/operator algebras**.

Extensions: Kondratiev, Mazzeo-Melrose, Schrohe, Schulze, Mitrea and Nistor, MK-Nistor, ...

L^2 -unique continuation property

Definition

Let $T : L^2(M; E) \rightarrow C_0^\infty(M; F)'$ be a continuous linear map. We say that T has **L^2 -unique continuation property** if

$$u \in L^2(M; E), Tu = 0 \text{ \& } u = 0 \text{ in an open subset of } M \implies u = 0 \text{ on } M.$$

Example

Let M be a connected Riemannian manifold.

- The exterior derivative $d : L^2(M) \rightarrow C_0^\infty(M, T^*M)'$ has the L^2 -unique continuation property.
- The deformation operator Def and the deformation Laplacian $\mathbf{L} := 2\text{Def}^*\text{Def}$ both satisfy the L^2 -unique continuation property.

Preliminaries

- Let $X \in \mathfrak{X}(M)$ = set of vector fields. The **deformation** of X (the *symmetric* part of ∇X):

$$\text{Def } X = \frac{1}{2} \mathcal{L}_X g,$$

where $\mathcal{L}_X$ is the Lie derivative in the direction of X .

- $\text{Def } X \in C^\infty(M, S^2 T^* M)$ the **deformation** operator:
 $\text{Def} : \mathfrak{X}(M) \rightarrow C^\infty(M, S^2 T^* M)$.
- If $\text{Def } X = 0$, then X is called a *Killing vector field*.

Lemma (Grosse, MK, Nistor, 2026)

Let M be a connected Riemannian manifold with straight cylindrical ends. Let X be a smooth *Killing* vector field on M that vanishes on a non-empty open subset of M . Then $X = 0$ (i.e. it vanishes on M).

Unique continuation and injectivity

Lemma

Let M be a connected Riemannian manifold with straight cylindrical ends or a *smooth, compact manifold*. Then the following statements hold.

- ① $\text{Def} \in \Psi_{\text{ess}}^1(M; TM, T^*M \otimes T^*M)$.
- ② The **deformation Laplacian** $\mathbf{L} := 2\text{Def}^*\text{Def} \in \Psi_{\text{ess}}^2(M; TM)$ is *elliptic*.
- ③ Def and Def^*Def have the L^2 -unique continuation property on M .

Gives the injectivity for $\mathbf{L} + V$

Theorem

If $V \in C^\infty(M) \cap \Psi_{\text{ess}}^0(M)$, $V \geq 0$ on M , and V is *non-zero at infinity*, then
 $\mathbf{L}_V := \mathbf{L} + V = 2\text{Def}^*\text{Def} + V : H^s(M; TM) \rightarrow H^{s-2}(M; TM)$
 is **invertible** for any $s \in \mathbb{R}$, and $\mathbf{L}_V^{-1} \in \Psi_{\text{ess}}^{-2}(M; T^*M)$.

Proof. There is no loss of generality to assume that M is connected.

- $\mathbf{L} = 2\text{Def}^*\text{Def}$ is in $\Psi_{\text{ess}}^2(M; TM)$ and is **elliptic**.

By assumption, we also have $V \in \Psi_{\text{ess}}^0(M) \cap C^\infty(M) \subset \Psi_{\text{ess}}^0(M; TM)$.
Thus, $\mathbf{L}_V = \mathbf{L} + V$ is **elliptic**.

Hence, if $\mathbf{L}_V$ is invertible on $L^2(M; TM)$, then $\mathbf{L}_V^{-1} \in \Psi_{\text{ess}}^{-2}(M; TM)$.

- In order to prove that $\mathbf{L}_V$ is invertible, we prove that
 - the indicial operator of $\mathbf{L}_V$, $\tilde{\mathbf{L}}_V$ is invertible on $H^2(\partial M_0, \times \mathbb{R}, TM)$,
 - use the **ellipticity** of $\mathbf{L}_V$.

Then the **Fredholm property** of $\mathbf{L}_V$ follows from the **Kondratiev-Melrose-Mendoza result**.

Then the Fredholm property of $\mathbf{L}_V$ and the L^2 -unique continuity property of $\mathbf{L} = 2\text{Def}^*\text{Def}$ imply the invertibility of $\mathbf{L}_V$ on $H^2(M, TM)$, and by regularity, on $H^s(M, TM)$ for any $s \in \mathbb{R}$. $\square$

Douglis-Nirenberg operators: jumps, ellipticity, and Fredholmness

Definition

Let M be a smooth manifold, let $E_i \rightarrow M$ smooth vector bundles, $1 \leq i \leq k$, and let $s_i, t_j \in \mathbb{R}$, $1 \leq i, j \leq k$. We set $\mathbf{s} := (s_1, \dots, s_k)$, $\mathbf{t} := (t_1, \dots, t_k)$, and $E := E_1 \oplus \dots \oplus E_k$. Then

$$\Psi^{[\mathbf{s}+\mathbf{t}]}(M; E) := \{P = [P_{ij}] \mid P_{ij} \in \Psi^{s_i+t_j}(M; E_j, E_i), 1 \leq i, j \leq k\}.$$

An operator $P = [P_{ij}] \in \Psi^{[\mathbf{s}+\mathbf{t}]}(M; TM \oplus \mathbb{C})$ is said to be of *Douglis-Nirenberg-order* $\leq [\mathbf{s} + \mathbf{t}]$. For $P = [P_{ij}] \in \Psi^{[\mathbf{s}+\mathbf{t}]}(M; TM \oplus \mathbb{C})$ let

$$\text{Symb}_{\mathbf{s}, \mathbf{t}}(P) := [\sigma_{s_i+t_j}(P_{ij})]$$

be its $(\mathbf{s}, \mathbf{t})$ -*principal symbol*, which is a suitable section of the lift of the endomorphism bundle $\text{End}(E)$ to $T^*M \setminus 0$. The operator P is said to be $(\mathbf{s}, \mathbf{t})$ -*Douglis-Nirenberg elliptic* if its $(\mathbf{s}, \mathbf{t})$ -principal symbol matrix $\text{Symb}_{\mathbf{s}, \mathbf{t}}(P)$ is invertible outside the zero section (i.e., on $T^*M \setminus 0$).

Example

Let $k = 2$, $\mathbf{s} = (s_1, s_2) = \mathbf{t} = (t_1, t_2) = (1, 0)$, $E_1 = TM$, and $E_2 = \mathbb{C}$. Then

$$\Xi := \Xi_{V, V_0} \in \Psi^{[\mathbf{s}+\mathbf{t}]}(M; TM \oplus \mathbb{C}) := \{P = [P_{ij}] \mid P_{ij} \in \Psi^{s_i+t_j}(M; E_j, E_i)\}.$$

Lemma

Let $\mathbf{s} = \mathbf{t} = (1, 0)$. Then the *generalized Stokes operator* $\Xi := \Xi_{V, V_0}$ belongs to $\Psi^{[\mathbf{s}+\mathbf{t}]}(M; TM \oplus \mathbb{C})$ and is *$(\mathbf{s}, \mathbf{t})$ -Douglis-Nirenberg-elliptic*:

$$\text{Symb}_{\mathbf{s}, \mathbf{t}}(\Xi)(\xi) = \begin{pmatrix} |\xi|^2 + \xi^\sharp \otimes \xi & \imath \xi^\sharp \\ -\imath \xi & -V_0 \end{pmatrix} \in \text{End}(TM \oplus \mathbb{C}),$$

which is *invertible* with inverse

$$\begin{pmatrix} \frac{1}{|\xi|^2} - \frac{V_0+1}{2V_0+1} \frac{1}{|\xi|^4} \xi^\sharp \otimes \xi & \frac{\imath}{(2V_0+1)|\xi|^2} \xi^\sharp \\ -\frac{\imath}{(2V_0+1)|\xi|^2} \xi & -\frac{2}{2V_0+1} \end{pmatrix} \in \text{End}(TM \oplus \mathbb{C}),$$

where $\sharp : T^*M \rightarrow TM$ is the isomorphism defined by the metric g of M .

The case of a manifold with straight cylindrical ends M

We let

$$\Psi_{\text{ess}}^{[\mathbf{s}+\mathbf{t}]}(M; E) := \{P = [P_{ij}] \mid P_{ij} \in \Psi_{\text{ess}}^{s_i+t_j}(M; E_j, E_i), 1 \leq i, j \leq k\}.$$

Let also $m = \max\{s_i + t_j\}$. Then $\Psi^{[\mathbf{s}+\mathbf{t}]}(M; E) \subset \Psi^m(M; E)$ and we have

$$\Psi_{\text{ess}}^{[\mathbf{s}+\mathbf{t}]}(M; E) = \Psi_{\text{ess}}^m(M; E) \cap \Psi^{[\mathbf{s}+\mathbf{t}]}(M; E).$$

Theorem

Let $M := M_0 \cup (\partial M_0 \times (-\infty, 0])$ be a manifold with straight cylindrical ends and $s_i, t_j \in \mathbb{R}$, $1 \leq i, j \leq k$, as before. Let $P = [P_{ij}] \in \Psi_{\text{ess}}^{[\mathbf{s}+\mathbf{t}]}(M; E)$ and $m \in \mathbb{R}$. Then the map

$$P : H^{m+[\mathbf{t}]}(M; E) := \bigoplus_{j=1}^k H^{m+t_j}(M; E_j) \rightarrow H^{m-[\mathbf{s}]}(M; E) := \bigoplus_{i=1}^k H^{m-s_i}(M; E_i)$$

is **Fredholm** if, and only if,

- (i) P is **Douglis-Nirenberg-elliptic** and
- (ii) the **limit operator** $\mathcal{R}_\infty(P) \in \Psi_{\text{ess}}^{[\mathbf{s}+\mathbf{t}]}(\partial M_0 \times \mathbb{R}; E)$ is **invertible**.

The Fredholm and invertibility properties of Ξ

We write $v \succ 0$ for a non-negative function v that does not vanish identically on any connected component of its domain.

Assumption

*Let $M := M_0 \cup (\partial M_0 \times (-\infty, 0])$ be a smooth, connected manifold with straight cylindrical ends with M_0 closed, non-empty and let $V \in \mathcal{C}_{\text{inv}}^\infty(M; TM)$ and $V_0 \in \mathcal{C}_{\text{inv}}^\infty(M)$ be our potentials. We shall say that V and V_0 satisfy the **positivity assumption at infinity on M** if*

$$\widehat{V} := \mathcal{R}_\infty V \succ 0 \quad \text{and} \quad \widehat{V}_0 := \mathcal{R}_\infty V_0 \succ 0 \quad \text{on } M.$$

The Moore-Penrose pseudoinverse of Ξ

Definition

Let $\mathcal{N}$ be the kernel of $\Xi : H^1(M; TM) \oplus L^2(M) \rightarrow H^{-1}(M; TM) \oplus L^2(M)$.

Assume that $\mathcal{N} \subset L^2(M; TM \oplus \mathbb{C})$ and that it is finite dimensional and let $p_{\mathcal{N}}$ be the L^2 -orthogonal projection onto $\mathcal{N}$. Let $\mathbf{s} = \mathbf{t} = (1, 0)$. Assume also that there exists a pseudodifferential operator

$$\Xi^{(-1)} \in \Psi_{\text{cl}}^{[-\mathbf{s}-\mathbf{t}]}(M; TM \oplus \mathbb{C}) := \begin{pmatrix} \Psi^{-2}(M; TM) & \Psi^{-1}(M; TM, \mathbb{C}) \\ \Psi^{-1}(M; \mathbb{C}, TM) & \Psi^0(M) \end{pmatrix} \quad (6.1)$$

such that

$$\begin{aligned} \Xi^{(-1)}(1 - p_{\mathcal{N}}) &= (1 - p_{\mathcal{N}})\Xi^{(-1)} = \Xi^{(-1)} \quad \text{and} \\ \Xi\Xi^{(-1)} &= \Xi^{(-1)}\Xi = 1 - p_{\mathcal{N}} \end{aligned} \quad (6.2)$$

i.e., we assume that Ξ is invertible on the orthogonal complement of its kernel $\mathcal{N}$.

The operator $\Xi^{(-1)}$ is uniquely determined by the relations (6.2) and will be called the **Moore-Penrose pseudoinverse** of $\Xi := \Xi_{V, V_0}$.

Theorem

Let M be a manifold with straight cylindrical ends and assume that $\widehat{V} \succ 0$ and $\widehat{V}_0 \succ 0$. Then the following properties hold.

- (i) The limit operator $\mathcal{R}_\infty(\Xi) \in \Psi^m(\partial M_0 \times \mathbb{R}; TM, \mathbb{C})$ is invertible.
- (ii) The generalized Stokes operator $\Xi := \Xi_{V, V_0}$ is Fredholm.
- (iii) Let $\mathbf{s} = \mathbf{t} = (1, 0)$. Consequently, the Moore-Penrose pseudoinverse of Ξ satisfies $\Xi^{(-1)} \in \Psi_{\text{ess}}^{[-\mathbf{s}, -\mathbf{t}]}(M; TM \oplus \mathbb{C})$.

We now use this result to prove the invertibility of Ξ under the following stronger assumption.

Assumption

We say that V and V_0 satisfy the **positivity assumption on M** if they satisfy the positivity assumption at infinity on M ([Assumption 19](#)) and $V \geq 0$ and $V_0 \geq 0$.

We can now prove our **main invertibility result** (Theorem 5.11 from [4]).

Theorem

Let M be a manifold with straight cylindrical ends and assume that V and V_0 satisfy the [positivity assumption on \$M\$](#) . Then the generalized Stokes operator $\Xi := \Xi_{V, V_0}$ is [invertible](#). Let $\mathbf{s} = \mathbf{t} = (1, 0)$. Consequently, its inverse satisfies $\Xi^{-1} \in \Psi_{\text{ess}}^{[-\mathbf{s}, -\mathbf{t}]}(M; TM \oplus \mathbb{C})$ and

$$\Xi := \Xi_{V, V_0} : H^{r+1}(M; TM) \oplus H^r(M) \rightarrow H^{r-1}(M; TM) \oplus H^r(M)$$

is an isomorphism for all $r \in \mathbb{R}$.

Definition of layer potentials

Recall that the **Moore-Penrose pseudoinverse $\Xi^{(-1)}$ of Ξ** is defined as the **inverse of the operator Ξ acting from the orthogonal of $\ker \Xi$ to its image.**

Lemma

Let M be a manifold with straight cylindrical ends. Let $\mathbf{s} = \mathbf{t} = (1, 0)$. We assume that V and V_0 satisfy the non-vanishing Assumption 19. Then

- (i) *The pseudoinverse $\Xi^{(-1)}$ has the form*

$$\Xi^{(-1)} =: \begin{pmatrix} \mathcal{A} & \mathcal{B} \\ \mathcal{C} & \mathcal{D} \end{pmatrix} \in \Psi_{\text{ess}}^{[-\mathbf{s}-\mathbf{t}]}(M; TM \oplus \mathbb{C}),$$

where $\mathcal{A} \in \Psi_{\text{ess}}^{-2}(M; TM)$, $\mathcal{C} = \mathcal{B}^ \in \Psi_{\text{ess}}^{-1}(M; TM, \mathbb{C})$, and $\mathcal{D} \in \Psi_{\text{ess}}^0(M)$.*

- (ii) *Consequently, we have $\sigma_{-2}(\mathcal{A})(x, \xi) = \frac{1}{|\xi|^2} - \frac{V_0+1}{2V_0+1} \frac{1}{|\xi|^4} \xi^\# \otimes \xi$, $\sigma_{-1}(\mathcal{B})(x, \xi) = \sigma_{-1}(\mathcal{C})(x, \xi)^* = \frac{2}{(2V_0+1)|\xi|^2} \xi^\#$, and $\sigma_0(\mathcal{D})(x, \xi) = -\frac{2}{2V_0+1}$.*

The existence of the pseudoinverse $\Xi^{(-1)}$ of Ξ allows us to define the single and double layer potential operators for the Stokes operator. We let $\Omega \subset M$ be a *compatible open subset* with straight cylindrical ends, that is, if $M = M_0 \cup [\partial M_0 \times (-\infty, 0)]$ then

$$\Omega \cap [\partial M_0 \times (-\infty, 0)] = \Omega' \times (-\infty, 0), \quad \Gamma := \partial \Omega. \quad (6.3)$$

Definition

Let $\mathbf{h} \in H^s(\Gamma; TM)$, $s \in \mathbb{R}$. The *single-layer potential* $\mathcal{S}_{\text{ST}}(\mathbf{h})$ for Ξ is then

$$\begin{pmatrix} \mathcal{V}_{\text{ST}}(\mathbf{h}) \\ \mathcal{P}_{\text{ST}}(\mathbf{h}) \end{pmatrix} := \mathcal{S}_{\text{ST}}(\mathbf{h}) := \Xi^{(-1)} \left[\begin{pmatrix} \mathbf{h} \\ 0 \end{pmatrix} \otimes \delta_\Gamma \right].$$

Similarly, the *double-layer potential* $\mathcal{D}_{\text{ST}}(\mathbf{h})$ for Ξ is given by:

$$\begin{pmatrix} \mathcal{W}_{\text{ST}}(\mathbf{h}) \\ \mathcal{Q}_{\text{ST}}(\mathbf{h}) \end{pmatrix} := \mathcal{D}_{\text{ST}}(\mathbf{h}) := \Xi^{(-1)} [\mathbf{T}_\nu^* (\mathbf{h} \otimes \delta_\Gamma)],$$

$$\langle \mathbf{h} \otimes \delta_\Gamma, \mathbf{f} \rangle := \int_\Gamma f(x) h(x) d\sigma(x).$$

Recall that

$$\begin{aligned} \mathbf{D}_\nu &: \mathcal{C}^\infty(M; TM) \rightarrow \mathcal{C}^\infty(M; TM), \\ \mathbf{D}_\nu X &:= \frac{1}{2}(\nu \otimes \sharp + \sharp \otimes \nu) \operatorname{Def}(X) = \langle \operatorname{Def}(X), \nu \otimes 1 \rangle^\sharp, \end{aligned} \quad (6.4)$$

$$\begin{aligned} \mathbf{T}_\nu &: \mathcal{C}^\infty(M; TM \oplus \mathbb{C}) \rightarrow \mathcal{C}^\infty(M; TM), \\ \mathbf{T}_\nu U &:= -2\mathbf{D}_\nu(\mathbf{u}) + p\nu, \end{aligned} \quad (6.5)$$

where $U := (\mathbf{u} \ p)^\top$ is a smooth, compactly supported section of $TM \oplus \mathbb{C}$, and

$$\mathbf{T}_\nu^*(\mathbf{u}) = \begin{pmatrix} -2\mathbf{D}_\nu^* \mathbf{u} \\ \nu \cdot \mathbf{u} \end{pmatrix}.$$

The inverse of the Stokes type operator Ξ_{V, V_0} Assumption on V and V_0

- Let $M = M_0 \cup (\partial M_0 \times (-\infty, 0])$ be a smooth manifold with straight cylindrical ends with ∂M_0 closed, non-empty and let $V \in C^\infty(M; TM)$, $V_0 \in C^\infty(M)$. We assume that V and V_0 satisfy the **positivity assumption on M** , i.e. that they satisfy the positivity assumption at infinity on M (Assumption 19) and $V \geq 0$ and $V_0 \geq 0$.

Theorem (MK-Nistor-Wendland, 2026)

Let M be a closed manifold or a *manifold with straight cylindrical ends*. Let us assume that V and V_0 satisfy the above non-vanishing assumption. Then the

Stokes operator $\Xi := \Xi_{V, V_0} = \Xi_{V, V_0} = \begin{pmatrix} \mathbf{L} + V & \nabla \\ \nabla^* & -V_0 \end{pmatrix}$ is *invertible* and

$$\Xi := \Xi_{V, V_0} : H^{r+1}(M; TM) \oplus H^r(M) \rightarrow H^{r-1}(M; TM) \oplus H^r(M)$$

is an *isomorphism* for all $r \in \mathbb{R}$.

Definition of layer potentials on M

Definition

Let $\mathbf{h} \in L^2(\partial\Omega; TM)$. The *single-layer potential* $\mathcal{S}_{\text{ST}}(\mathbf{h})$, the *velocity single-layer potential* $\mathcal{V}_{\text{ST}}(\mathbf{h})$, and the *pressure single-layer potential* $\mathcal{Q}_{\text{ST}}^s(\mathbf{h})$ for Ξ are defined by:

$$\mathcal{S}_{\text{ST}}(\mathbf{h}) := \begin{pmatrix} \mathcal{V}_{\text{ST}}(\mathbf{h}) \\ \mathcal{Q}_{\text{ST}}^s(\mathbf{h}) \end{pmatrix} := \Xi^{-1} \left[\begin{pmatrix} \mathbf{h} \\ 0 \end{pmatrix} \otimes \delta_{\partial\Omega} \right].$$

Similarly, the *double-layer potential* $\mathcal{D}_{\text{ST}}(\mathbf{h})$, the *double-layer velocity potential* $\mathcal{W}_{\text{ST}}(\mathbf{h})$, and the *double-layer pressure potential* $\mathcal{Q}_{\text{ST}}(\mathbf{h})$ for Ξ are defined by:

$$\mathcal{D}_{\text{ST}}(\mathbf{h}) := \begin{pmatrix} \mathcal{W}_{\text{ST}}(\mathbf{h}) \\ \mathcal{Q}_{\text{ST}}^d(\mathbf{h}) \end{pmatrix} := \Xi^{-1} [\mathbf{T}_\nu^* (\mathbf{h} \otimes \delta_{\partial\Omega})].$$

The double-layer potentials $\mathcal{W}_{\text{ST}}(\mathbf{h})$ and $\mathcal{Q}_{\text{ST}}^d(\mathbf{h})$ satisfy (with $L = 2\text{Def}^*\text{Def}$)

$$\begin{cases} (\mathbf{L} + \mathbf{V})\mathcal{W}_{\text{ST}}(\mathbf{h}) + \nabla \mathcal{Q}_{\text{ST}}^d(\mathbf{h}) = 0 & \text{in } M \setminus \partial\Omega \\ \nabla^* \mathcal{W}_{\text{ST}}(\mathbf{h}) - \mathbf{V}_0 \mathcal{Q}_{\text{ST}}^d(\mathbf{h}) = 0 & \text{in } M \setminus \partial\Omega. \end{cases}$$

We have the explicit form of these layer potentials:

$$\begin{aligned}\mathcal{W}_{\text{ST}}(\mathbf{h}) &= \begin{pmatrix} \mathcal{A} & \mathcal{B} \end{pmatrix} \begin{pmatrix} -2\mathbf{D}_{\nu}^* \\ \nu^{\sharp} \end{pmatrix} (\mathbf{h} \otimes \delta_{\partial\Omega}) = (-2\mathcal{A}\mathbf{D}_{\nu}^* + \mathcal{B}\nu^{\sharp}) (\mathbf{h} \otimes \delta_{\partial\Omega}) \\ \mathcal{Q}_{\text{ST}}^d(\mathbf{h}) &= \begin{pmatrix} \mathcal{C} & \mathcal{D} \end{pmatrix} \begin{pmatrix} -2\mathbf{D}_{\nu}^* \\ \nu^{\sharp} \end{pmatrix} (\mathbf{h} \otimes \delta_{\partial\Omega}) = (-2\mathcal{C}\mathbf{D}_{\nu}^* + \mathcal{D}\nu^{\sharp}) (\mathbf{h} \otimes \delta_{\partial\Omega}), \\ \mathcal{S}_{\text{ST}}(\mathbf{h}) &= \begin{pmatrix} \mathcal{A}(\mathbf{h} \otimes \delta_{\partial\Omega}) & \mathcal{C}(\mathbf{h} \otimes \delta_{\partial\Omega}) \end{pmatrix}^{\top}.\end{aligned}$$

Lemma

Let M be a manifold with straight cylindrical ends and that $\Omega \subset M$ is a compatible open subset (Equation (6.3)). Let $\Omega_+ := \Omega$, $\Omega_- := M \setminus \overline{\Omega}$, and $\Gamma := \partial\Omega$ be as before. Let $s \in \mathbb{R}$ and $\mathbf{h} \in H^s(\Gamma; TM)$. Then

$$\mathcal{V}_{\text{ST}}(\mathbf{h})|_{\Omega_{\pm}} \in H^{s+\frac{3}{2}}(\Omega_{\pm}; TM) \text{ and } \mathcal{P}_{\text{ST}}(\mathbf{h})|_{\Omega_{\pm}} \in H^{s+\frac{1}{2}}(\Omega_{\pm}).$$

Similarly, let $s \in \mathbb{R}$ and $\mathbf{h} \in H^s(\Gamma; TM)$. Then

$$\mathcal{W}_{\text{ST}}(\mathbf{h})|_{\Omega_{\pm}} \in H^{s+\frac{1}{2}}(\Omega_{\pm}; TM) \text{ and } \mathcal{Q}_{\text{ST}}(\mathbf{h})|_{\Omega_{\pm}} \in H^{s-\frac{1}{2}}(\Omega_{\pm}).$$

Definition

Let $\mathcal{A}, \mathcal{B}$ be as in the definition of layer potentials. We then let

- $\mathbf{S} := \mathcal{A}_0 \in \Psi_{\text{ess}}^{-1}(\partial\Omega; TM)$, that is, the restriction of $\mathcal{A}$ to $\partial\Omega$.
- $\mathbf{P} := -2\mathcal{A}\mathbf{D}_{\nu}^* + \mathcal{B}\nu^{\sharp} \in \Psi_{\text{ess}}^{-1}(M; TM)$ (the pseudodifferential operator defining the vector part $\mathcal{D}_{\text{ST}}$ of the double-layer potential).
- $\mathbf{K} := \mathbf{P}_0 := (-2\mathcal{A}\mathbf{D}_{\nu}^* + \mathcal{B}\nu^{\sharp})_0 \in \Psi_{\text{ess}}^0(\partial\Omega; TM)$ is the restriction of $\mathbf{P} := -2\mathcal{A}\mathbf{D}_{\nu}^* + \mathcal{B}\nu^{\sharp}$ to $\partial\Omega$.

The jump relations for the double-layer potentials

Let g_+ (g_-) be the interior (exterior) *normal limits* at $\partial\Omega$ of a function $g : M \setminus \partial\Omega \rightarrow \mathbb{R}$, when they exist.

Theorem

Let $\mathbf{K} := \mathbf{P}_0 \in \Psi_{\text{ess}}^0(\partial\Omega; TM)$ be as above. Then

$$\mathcal{W}_{\text{ST}}(\mathbf{h})_{\pm} = \left[\pm \frac{1}{2} \mathbb{I} + \mathbf{K} \right] \mathbf{h},$$

where $\sigma_0(\mathbf{K}; \xi') = \frac{iV_0}{2(2V_0+1)|\xi'|} (\boldsymbol{\nu} \otimes \xi' - \xi'^{\#} \otimes \boldsymbol{\nu}^{\#})$. In particular, the two operators $\pm \frac{1}{2} \mathbb{I} + \mathbf{K}$ are *elliptic and have self-adjoint principal symbols*.

The invertibility of the single-layer potential operator

Lemma

We have the equality $\left(\frac{1}{2}\mathbb{I} + \mathbf{K}\right)\mathbf{S} = \mathbf{S}\left(\frac{1}{2}\mathbb{I} + \mathbf{K}^*\right) \in \Psi_{\text{ess}}^{-1}(\partial\Omega; TM)$.

Theorem

Let M be a connected manifold that is either closed or with straight cylindrical ends. Assume that M , V , and V_0 satisfy the non-vanishing Assumptions 32 and let $\mathbf{S} \in \Psi_{\text{ess}}^{-1}(\partial\Omega; TM)$ be the single-layer potential operator. Then $\mathbf{S}$ is invertible, $\mathbf{S}^{-1} \in \Psi_{\text{ess}}^1(\partial\Omega; TM)$, and hence $\mathbf{S} : H^s(\partial\Omega; TM) \rightarrow H^{s+1}(\partial\Omega; TM)$ is an *isomorphism* for all $s \in \mathbb{R}$.

The invertibility of the double-layer potential operator $\frac{1}{2}\mathbb{I} + K$

Theorem (MK-Nistor-Wendland, 2026)

Let M be a connected manifold that is either *closed* or *with straight cylindrical ends*. Assume that M , V , and V_0 satisfy the non-vanishing Assumption 32, and that Ω is connected. Then $[\frac{1}{2}\mathbb{I} + K] \in \Psi_{\text{ess}}^0(\partial\Omega; TM)$ is invertible and the operator $\frac{1}{2}\mathbb{I} + K : H^s(\partial\Omega; TM) \rightarrow H^s(\partial\Omega; TM)$ is *an isomorphism* for all $s \in \mathbb{R}$.

Proof. There are several steps based on: **the Fredholm property, the properties of $\Psi_{\text{ess}}^m(M; TM)$, jump relations.**

- The operator $\frac{1}{2}\mathbb{I} + K : L^2(\partial\Omega; TM) \rightarrow L^2(\partial\Omega; TM)$ is Fredholm of index zero and injective, thus, invertible.
- The operators $\frac{1}{2}\mathbb{I} + K : H^s(\partial\Omega; TM) \rightarrow H^s(\partial\Omega; TM)$, $\frac{1}{2}\mathbb{I} + K^* : H^{-s}(\partial\Omega; TM) \rightarrow H^{-s}(\partial\Omega; TM)$ are invertible for any $s \geq 0$.
- Duality yields the invertibility for any $s \leq 0$, and hence for any $s \in \mathbb{R}$. $\square$

Well-posedness of the Dirichlet problem

Theorem

Let $M = M_0 \cup [M' \times (-\infty, 0)]$ be a manifold with straight cylindrical ends, $M' = \partial M_0$, let $\Omega \subset M$ be a smooth domain with straight cylindrical ends that is on one side of its boundary $\Gamma := \partial\Omega \neq \emptyset$, and let $m \in \mathbb{Z}_+$. Let $V \in C_{\text{inv}}^\infty(M; TM)$ and $V_0 \in C_{\text{inv}}^\infty(M)$ be such that:

- (i) $V \geq 0$ and $V_0 \geq 0$ on M ; $V_0 \succ 0$ on Ω ,
- (ii) $\mathcal{R}_\infty(V_0) \succ 0$ on Ω' , and if $\Omega_0 \subseteq \Omega'$ is a connected component of M' , then $\mathcal{R}_\infty(V) \not\equiv 0$ on Ω_0 .

Then, there is $C_m \geq 0$ s.t., for any $\mathbf{f} \in H^{m+1/2}(\Gamma; TM)$, the **Dirichlet problem**

$$\equiv U := \begin{pmatrix} L + V & \nabla \\ \nabla^* & -V_0 \end{pmatrix} U = 0 \text{ in } \Omega, \text{ and } \mathbf{u}|_{\partial\Omega} = \mathbf{f},$$

has a **unique solution** $U = (\mathbf{u} \ p)^\top \in H^{m+1}(\Omega; TM) \oplus H^m(\Omega)$ and

$$\|\mathbf{u}\|_{H^{m+1}(\Omega; TM)} + \|p\|_{H^m(\Omega)} \leq C_m \|\mathbf{f}\|_{H^{m+1/2}(\Gamma; TM)}.$$

The existence part of the proof

- The solution problem has the **double-layer integral representation**

$$(\mathbf{u} \ p)^\top = \mathcal{D}_{\text{ST}} \left(\frac{1}{2} \mathbb{I} + \mathbf{K} \right)^{-1} \mathbf{f} = \begin{pmatrix} \mathcal{W}_{\text{ST}} \left(\frac{1}{2} \mathbb{I} + \mathbf{K} \right)^{-1} \mathbf{f} \\ \mathcal{Q}_{\text{ST}}^d \left(\frac{1}{2} \mathbb{I} + \mathbf{K} \right)^{-1} \mathbf{f} \end{pmatrix}.$$

- In terms of the **single-layer potential approach**

$$(\mathbf{u} \ p)^\top = \mathcal{S}_{\text{ST}} (\mathbf{S}^{-1} \mathbf{f}).$$

- Uniqueness follows from Green's formula.

The boundary behavior of $\mathbf{T}_\nu \mathcal{D}_{\text{ST}}$

- For $\mathbf{f} \in H^{3/2}(\partial\Omega; TM)$ arbitrary, let $\mathbf{U} := (\mathbf{u} \ p)^\top \in H^2(\Omega; TM) \oplus H^1(\Omega)$ be the unique solution of the Dirichlet problem

$$\Xi_{V, V_0} \mathbf{U} = 0 \text{ on } \Omega, \quad \mathbf{u}|_{\partial\Omega} = \mathbf{f}. \quad (9.1)$$

Then the *Dirichlet-to-Neumann map* $\mathcal{N}_{\text{ST}} : H^{3/2}(\partial\Omega; TM) \rightarrow H^{1/2}(\partial\Omega; TM)$:

$$\mathcal{N}_{\text{ST}} \mathbf{f} = [\mathbf{T}_\nu \mathbf{U}]_+ \text{ on } \partial\Omega. \quad (9.2)$$

The next result shows that there is no jump of the conormal derivative $\mathbf{T}_\nu \mathcal{D}_{\text{ST}}(\mathbf{f})$ across $\partial\Omega$ (see Proposition 11.4, Taylor (1996) for Laplacian).

Theorem

$$[\mathbf{T}_\nu \mathcal{D}_{\text{ST}}(\mathbf{f})]_+ = [\mathbf{T}_\nu \mathcal{D}_{\text{ST}}(\mathbf{f})]_- = \left(\frac{1}{2}\mathbb{I} + \mathbf{K}^*\right) \mathcal{N}_{\text{ST}} \mathbf{f}.$$

Application: Well-posedness of Dirichlet problem for the Navier-Stokes system

Theorem

Let M , V , and V_0 satisfy the assumptions in Theorem 35. Let $\Omega \subset M$ be a smooth domain with straight cylindrical ends in the smooth connected manifold with straight cylindrical ends M . Assume also that Ω is the interior of $\overline{\Omega}$. If $m \in \mathbb{Z}_+$ and $2 \leq \dim M = n < 2(m+2)$, then there exist $\zeta, \eta \in (0, \infty)$ such that for all $\mathbf{f} \in H^{m+1/2}(\Gamma; TM)$ and $\mathbf{h} \in H^{m-1}(\Omega; TM)$ such that

$$\|\mathbf{h}\|_{H^{m-1}(\Omega; TM)} + \|\mathbf{f}\|_{H^{m+1/2}(\Gamma; TM)} \leq \zeta, \quad (9.3)$$

the *Dirichlet problem* for the *Navier-Stokes system*

$$\equiv U + \begin{pmatrix} \nabla_{\mathbf{u}} \mathbf{u} \\ 0 \end{pmatrix} = \begin{pmatrix} \mathbf{h} \\ 0 \end{pmatrix} \text{ in } \Omega \text{ and } \mathbf{u}|_{\Gamma} = \mathbf{f} \quad (9.4)$$

has a unique solution $U = (\mathbf{u} \ p)^{\top} \in H^{m+1}(\Omega; TM) \oplus H^m(\Omega)$ such that

$$\|\mathbf{u}\|_{H^{m+1}(\Omega; TM)} \leq \eta. \quad (9.5)$$

The existence part of the proof

Lemma

Let M be as above and $\Omega \subset M$ be a smooth domain with straight cylindrical ends in the connected manifold with straight cylindrical ends M . Assume that $m \in \mathbb{Z}_+$ and that $\dim M = n < 2(m+2)$. Then there is $C > 0$, such that

$$\|\nabla_{\mathbf{u}} \mathbf{v}\|_{H^{m-1}(\Omega; TM)} \leq C \|\mathbf{u}\|_{H^{m+1}(\Omega; TM)} \|\mathbf{v}\|_{H^{m+1}(\Omega; TM)},$$

$$\|\nabla_{\mathbf{u}} \mathbf{u} - \nabla_{\mathbf{v}} \mathbf{v}\|_{H^{m-1}(\Omega; TM)} \leq C \left(\|\mathbf{u}\|_{H^{m+1}(\Omega; TM)} + \|\mathbf{v}\|_{H^{m+1}(\Omega; TM)} \right) \|\mathbf{u} - \mathbf{v}\|_{H^{m+1}(\Omega; TM)},$$

for all $\mathbf{u}, \mathbf{v} \in H^{m+1}(\Omega; TM)$.

- Existence of a *fixed point of a non-linear operator*, which is a solution of the non-linear problem.

Conclusion

- The **well-posedness of the Dirichlet problem** for the generalized Stokes operator via a layer potential approach - essentially translation invariant pseudodifferential operators.
- The **Neumann problem** on domains with conical points/cylindrical ends behaves **quite differently** from the Dirichlet problem (joint work with **M. Lanza de Cristoforis and V. Nistor**); M. Costabel, M. Dauge, V. Mazya, V. Nistor, ...
- Extensions to other **non-compact manifolds**, in progress (joint work with V. Nistor, C. Anghel-Stan, ...)



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Thank you for your attention !